

Crystal

A naturally occurring solid substance whose atoms are arranged in a regular three dimensions pattern.

Decay

A process in which unstable atoms of radioactive elements break apart. The atoms of one element are transformed into those of another element during decay.

Density

The amount of matter held within a known volume of a material.

Dissolve

To become completely mixed into another substance. In most cases, a solid, such as salt, dissolves in a liquid, such as water.

Earthenware

A basic form of pottery, in which clay is heated to make it form hard structures.

Electrolysis

A process in which a compound is split into simpler substances using an electric current. Many elements, such as aluminium, are purified from their compounds in ores in this way.

Electrode

An electrical contact in an electric circuit. Electrodes can have a positive or negative charge.

Electron

A negatively charged particle inside an atom. Electrons orbit the atom's nucleus (or core) in layers called shells. They are also exchanged or shared by atoms to make bonds that hold molecules together.

Element

A pure substance that cannot be broken down into any simpler substances. Elements are the building blocks of matter. There are 118 known elements on Earth.

Expand

To make or become bigger. Solids, liquids, and gases usually expand when the temperature rises.

Fission

A process in which the nucleus of an unstable atom splits in two when it collides with a neutron. It releases more neutrons that start the cycle again, and this continues as a chain reaction. Many unstable nuclei undergo fission spontaneously, without being hit by a neutron. Fission releases a lot of energy. The process is used to generate electricity in nuclear power plants, and it can trigger atom bomb explosions.

Flammable

A way of describing a material that can catch fire easily.

Fusion

A process in which small atoms, such as those of hydrogen, are fused together with such force that they merge to form larger atoms, while releasing a lot of energy. The Sun is powered by the fusion of hydrogen atoms into helium in its central core.

Gas

A state in which the particles of matter (atoms or molecules) are not attached to each other, and move freely. A gas can flow, take any shape, and fill any container.

Group

A set of elements in a column in the periodic table. Elements in a group have similar properties because each atom has the same number of electrons in its outer shell.

Halogen

A member of a group near the far right of the periodic table. Halogens form salts with metals. They are reactive, non-metallic elements.

Hardness

A measure of how easy it is to scratch or cut a substance with another substance.

Hydroxide

A type of compound containing hydrogen and oxygen, and normally a metallic element.

This is a chunk of ytterbium, a **lanthanide**.

**Ion**

An atom or a group of atoms that have an electric charge. While atoms have no overall charge, they become positive ions if they lose electrons or negative ions if they gain electrons.

Insulator

A substance that does not let heat or electricity flow easily through it.

Isotope

A form of an element with atoms that have the same number of protons but a different number of neutrons.

Lanthanide

A member of a set of metals with relatively large atoms. The elements in this series sit with the actinides, below the main part of the periodic table.



This **element** calcium is in crystal form.